Two- and three-recorder candidate signals, top 15 in order of approximate likelihood ratio (none favoured over noise)

Rank by ln LR	ln LR	Rank by largest p	Recorders	Arrival times (UTC)	Detector and band	p-values (block-permutation p, panels)	Time differences, s: observed [predicted 99 % range]	Arc point from the time difference; impact time (UTC)	Coupling for detection at all recorders with probability 0.5 (× reference crash, provisional)	Sound direction at Cape Leeuwin against Kadri	Searches label
1	-0.64	771	Diego Garcia S. H08S × Perth Canyon 3315	01:02:53; 00:47:52	matched filter 1-9 Hz; band energy 2-5 Hz	0.07 (0.12); 0.12	-900 [-1573, -698]	36.9°S 90.0°E; 00:21:21	0.87 (4,147×)	n/a (no Cape Leeuwin signal)	screen
2	-2.51	3239	Diego Garcia S. H08S × Perth Canyon 3315	01:03:13; 00:47:52	STA/LTA 1-9 Hz; band energy 2-5 Hz	0.19 (0.29); 0.12	-921 [-1573, -698]	36.7°S 90.4°E; 00:21:44	0.55 (2,617×)	n/a (no Cape Leeuwin signal)	screen
3	-2.73	7890	Diego Garcia S. H08S × Perth Canyon 3315	01:02:53; 00:47:16	matched filter 1-9 Hz; band energy 2-5 Hz	0.07 (0.12); 0.24	-936 [-1573, -698]	36.5°S 90.6°E; 00:21:25	0.38 (1,810×)	n/a (no Cape Leeuwin signal)	screen
4	-2.83	12618	Diego Garcia S. H08S × Perth Canyon 3315	01:04:52; 00:49:48	STA/LTA 1-9 Hz; matched filter 5-40 Hz	0.28 (0.78); 0.22	-905 [-1573, -698]	36.8°S 90.1°E; 00:23:21	0.056 (268×)	n/a (no Cape Leeuwin signal)	screen
5	-2.95	1778	Diego Garcia S. H08S × Perth Canyon 3376 × Portland 3274	01:07:22; 00:52:11; 01:15:51	band energy 5-9 Hz; matched filter 5-40 Hz; band energy 5-10 Hz	0.16 (0.19); 0.17; 0.086	H08S-3376 -911 [-1573,-696] H08S-3274 509 [86,740] 3376-3274 1420 [1310,1711]	36.7°S 90.2°E; 00:25:52	0.71 (3,371×)	n/a (no Cape Leeuwin signal)	screen
6	-3.03	6361	Diego Garcia S. H08S × Perth Canyon 3376 × Portland 3274	01:07:02; 00:52:11; 01:15:51	STA/LTA 1-9 Hz; matched filter 5-40 Hz; band energy 5-10 Hz	0.22 (0.53); 0.17; 0.086	H08S-3376 -890 [-1573,-696] H08S-3274 530 [86,740] 3376-3274 1420 [1310,1711]	37.0°S 89.9°E; 00:25:28	0.54 (2,557×)	n/a (no Cape Leeuwin signal)	screen
7	-3.42	6725	Diego Garcia S. H08S × Perth Canyon 3315	01:04:23; 00:49:48	band energy 5-9 Hz; matched filter 5-40 Hz	0.22 (0.5); 0.22	-875 [-1573, -698]	37.2°S 89.6°E; 00:22:48	0.17 (790×)	n/a (no Cape Leeuwin signal)	screen
8	-3.48	1778	Diego Garcia S. H08S × Perth Canyon 3376 × Portland 3274	01:07:52; 00:52:11; 01:15:51	residual envelope; matched filter 5-40 Hz; band energy 5-10 Hz	0.094 (0.094); 0.17; 0.086	H08S-3376 -941 [-1573,-696] H08S-3274 479 [86,740] 3376-3274 1420 [1310,1711]	36.4°S 90.7°E; 00:26:25	0.81 (3,871×)	n/a (no Cape Leeuwin signal)	screen
9	-3.54	423	Perth Canyon 3376 × Portland 3275	00:55:01; 01:20:35	band energy 2-5 Hz; matched filter 5-40 Hz	0.096; 0.0082	1534 [1308, 1710]	33.5°S 94.6°E; 00:33:23	0.18 (867×)	n/a (no Cape Leeuwin signal)	mark
10	-3.58	6523	Diego Garcia S. H08S × Perth Canyon 3315	01:03:23; 00:47:52	band energy 5-9 Hz; band energy 2-5 Hz	0.22 (0.49); 0.12	-931 [-1573, -698]	36.5°S 90.5°E; 00:21:55	0.76 (3,612×)	n/a (no Cape Leeuwin signal)	screen
11	-3.66	1778	Diego Garcia S. H08S × Perth Canyon 3376	01:07:22; 00:52:11	band energy 5-9 Hz; matched filter 5-40 Hz	0.16 (0.19); 0.17	-911 [-1573, -696]	36.7°S 90.2°E; 00:25:52	0.16 (772×)	n/a (no Cape Leeuwin signal)	screen
12	-3.66	13407	Diego Garcia S. H08S × Perth Canyon 3376 × Portland 3274	01:07:22; 00:52:11; 01:15:54	band energy 5-9 Hz; matched filter 5-40 Hz; band energy 10-20 Hz	0.16 (0.19); 0.17; 0.28	H08S-3376 -911 [-1573,-696] H08S-3274 512 [86,740] 3376-3274 1423 [1310,1711]	36.7°S 90.2°E; 00:25:52	0.71 (3,371×)	n/a (no Cape Leeuwin signal)	screen
13	-3.69	1778	Perth Canyon 3376 × Portland 3274	00:52:11; 01:15:51	matched filter 5-40 Hz; band energy 5-10 Hz	0.17; 0.086	1420 [1310, 1711]	36.8°S 90.1°E; 00:25:42	0.47 (2,227×)	n/a (no Cape Leeuwin signal)	screen
14	-3.75	13407	Diego Garcia S. H08S × Perth Canyon 3376 × Portland 3274	01:07:02; 00:52:11; 01:15:54	STA/LTA 1-9 Hz; matched filter 5-40 Hz; band energy 10-20 Hz	0.22 (0.53); 0.17; 0.28	H08S-3376 -890 [-1573,-696] H08S-3274 533 [86,740] 3376-3274 1423 [1310,1711]	37.0°S 89.9°E; 00:25:28	0.54 (2,557×)	n/a (no Cape Leeuwin signal)	screen
15	-3.79	6607	Diego Garcia S. H08S × Perth Canyon 3315	01:04:12; 00:49:48	matched filter 1-9 Hz; matched filter 5-40 Hz	0.22 (0.55); 0.22	-864 [-1573, -698]	37.3°S 89.4°E; 00:22:36	0.042 (199×)	n/a (no Cape Leeuwin signal)	screen
146	-6.23	54	Cape Leeuwin H01W × Diego Garcia S. H08S	00:52:04; 01:12:49	matched filter 1-9 Hz; matched filter 1-9 Hz	0.043 (0.068); 0.039 (0.3)	1245 [809, 1536]	33.6°S 94.5°E; 00:31:41	0.069 (329×)	inconsistent: predicted 269°, nearest |Δ| 37°	report

(i) Each row is a set of possible signals at two or three recorders whose times agree with one simulated impact. The rank uses an approximate likelihood ratio: how likely the signals are if they come from the aircraft, compared with how often noise gives such signals. ln LR below 0 means noise explains the set better. All sets have ln LR below 0. The last row (shaded) is the pair the searches named as the strongest two-recorder pair; its sound direction does not agree with the directions Kadri published. The coupling is the share of impact energy that must become sound; above 1 is not possible. Estimates only, no detection. Rank by largest p is over all 15,652 sets.

(ii) EXPLORATORY ranking, pre-registered at module commit 9f47cf7 (prepare/ranked_candidates/prereg.py); not part of the frozen searches. Source posterior run J geometry sample, 00:19 Held Out (D0'), p = 0.5. ln LR = ln sum_i w_i mean_eta prod_c[P_D(snr; alpha = p_c) K_t A G] - ln prod_c(p_c/55 s); eta log-uniform [1e-6, 1e-2]; P_D block 7 fits (IMS rows: IMOS-proxy noise + 4.8 dB); TL block 11 machinery with the corrected IMOS RAM-grid table; A = level consistency (sigma 10 dB); G = Kadri Table 1 bearing within 30 s, t3 scale 3.3 deg. Arc point: crossing of the observed lag on the 7th-arc grid (block 11 method; the crossing with most D0' mass within +/-1 deg). Coupling: eta for prod P_D = 0.5 at alpha 0.05 with run J median E and tau within +/-1 deg; reference F-35A eta 2.1e-4 (provisional - from Brown 2026's quote of Metz 2023; re-derivation from Metz pending). Star pair: locus bearing 268.9 deg from H01W (1,812 km), 144.8 deg from H08S (3,660 km); Kadri gives 57 or 306.18 deg for the 00:52 signal (text p. 9 vs Table 1 caption), neither within 37 deg; nearest Table 1 bearing 00:52:50 at 274.84 deg, 46 s later. H08S raw-panel rows are airgun-dominated (G-S9); Perth Canyon 3315 2-5 Hz rows are below the 8 Hz hardware corner and use the 3376 null (G-S4). Digitised Kadri (2024) Fig. 9 panels; text time 00:52:03, caption time 00:54:30. Full tables: ranked_multisite.csv (15,652 sets: 3,321 pairs + 12,331 triples; 11,067 triples, including every IMOS x IMOS x IMOS set, contain blocked Scott Reef 3250 and 63 more have no simulated impact that fits all three times: ln LR -inf). PREDICTIVE, NOT CALIBRATED; C1, C2.